

ENVS 193DS Homework #3

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Setting up

```
# reading in package: tidyverse
library(tidyverse)
# reading in package: here
library(here)
# reading in package: janitor
library(janitor)
# reading in package: readxl
library(readxl)
```

```
# storing read kelp data as "kelp" object
kelp <- read.csv("data/temp-kelp.csv")
```

```
# storing read personal data as "personal_data" object
personal_data <- read.csv("data/personal_data(Sheet1).csv")
```

Problem 1: Giant kelp fronds

a. Appropriate tests

The appropriate tests to determine the strength of the relationship between temperature and giant kelp frond elongation rate are a Pearson correlation and a simple linear regression. A Pearson correlation evaluates the strength and direction of the linear relationship between two variables; in this case, ocean temperature and giant kelp frond elongation rate. A simple linear regression also evaluates the relationship between two variables, but it specifically shows elongation rate as a response variable that is predicted by temperature. Creating a linear model for this data can show how much growth changes per 1 degree Celsius increase in temperature.

b. Creating a visualization

```
# creating new object called "kelp_lm" out of linear model evaluating
↪ relationship between kelp elongation rate and temperature in deg celsius
kelp_lm <- lm(kelp_elong ~ temp_c, data = kelp) # kelp data frame
```

```

# creating object kelp_preds by combining (cbind) the original kelp data set
↪ and predicted values and confidence intervals from the linear model above
↪
kelp_preds <- cbind(
  # original data set
  kelp,
  # generating predicted values from linear model
  predict(kelp_lm, interval = "confidence") # interval = "confidence"
  ↪ calculates lower and upper confidence intervals and predicted
  ↪ regression line values
)

```

```

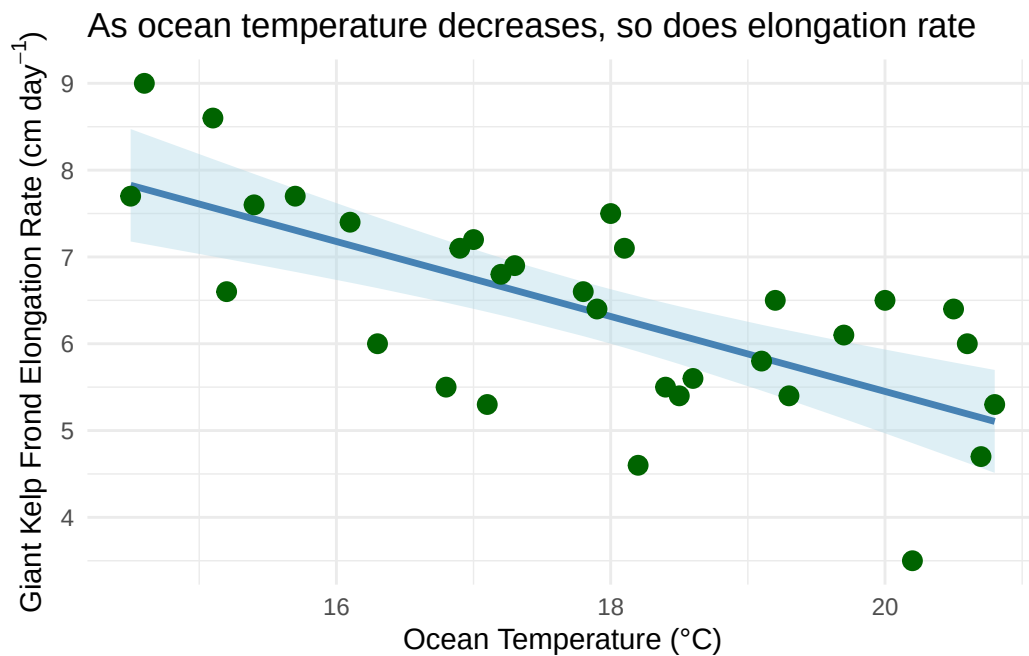
# base layer ggplot with kelp_preds data frame
ggplot(data = kelp_preds,
  # setting aesthetics
  mapping = aes(
    # x is temperature measured in celsius
    x = temp_c,
    # y is giant kelp frond elongation rate measured in cm per day
    y = kelp_elong)) +
# first layer geom_ribbon to visually show confidence intervals for linear
↪ model
geom_ribbon(aes
  # setting the ribbon bounds to be the lower and upper
  ↪ confidence intervals
  (ymin = lwr, ymax = upr),
  # making our ribbon a nice light blue color
  fill = "lightblue",
  # making it so that you can see through our ribbon
  alpha = 0.4) +
# second layer geom_line to show relationship line
geom_line(aes
  # uses predicted values previously generated from lm as y-values
  ↪ for our line
  (y = fit),
  # color that compliments our ribbon
  color = "steelblue",
  # adjusting width of the line
  linewidth = 1.2) +
# third layer geom_point to show individual observations in pretty green
↪ color
geom_point(color = "darkgreen",

```

```

# making points noticably large
size = 3) +
# using labs to set axis titles
labs(
  # x axis: ocean temperature
  x = "Ocean Temperature (°C)",
  # y axis: giant kelp frond elongation rate
  y = expression("Giant Kelp Frond Elongation Rate (" * cm ~ day^{-1} *
    ↵ ")"),
  # visualization title: finding from visualization
  title = "As ocean temperature decreases, so does elongation rate"
) +
# setting a nice, basic theme
theme_minimal()

```



c. Checking assumptions and running a test

Running tests:

```
# creating new object called "kelp_lm" out of linear model evaluating
↪ relationship between kept elongation rate and temperature in deg celsius
↪ (done in previous part, but wanted to show again for this problem part)
kelp_lm <- lm(kelp_elong ~ temp_c, data = kelp) # data frame kelp

# display summary results of kelp linear model
summary(kelp_lm)
```

Call:

```
lm(formula = kelp_elong ~ temp_c, data = kelp)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.8643	-0.5017	0.1790	0.5659	1.2177

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.08645	1.49219	9.440	1.72e-10 ***
temp_c	-0.43179	0.08321	-5.189	1.37e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom

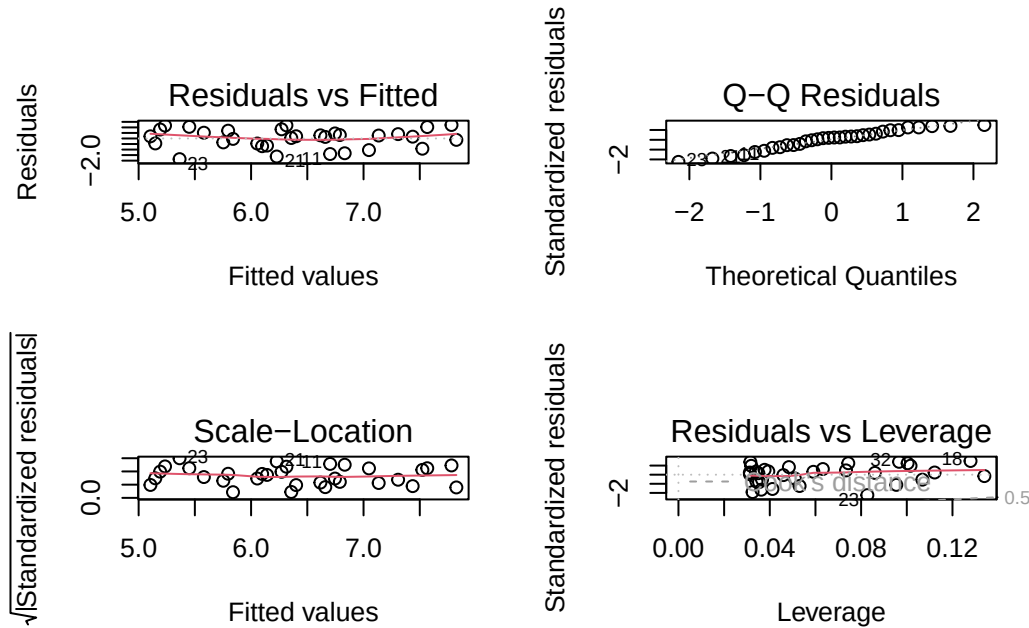
Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 1 and 30 DF, p-value: 1.366e-05

Note: I decided that showing my code in which I ran my test is appropriate in this situation because this same code is used in the next part to check my assumptions.

Checking assumptions:

```
# creating diagnostic plots to check model assumptions
par(mfrow = c(2, 2))
# plot assumption plots: residuals vs fitted, q-q residuals, scale-location,
↪ and residuals vs leverage
plot(kelp_lm)
```



Here, I checked the assumptions of the linearity of data, the normality of residuals, homoscedasticity (constant variance), and if there are outliers significant enough to influence our plot.

The residuals vs fitted and scale-location plots featured points that are somewhat evenly and randomly distributed around the horizontal dotted line, indicating homoscedasticity. The Q-Q residuals plot suggests normally enough distributed residuals. Finally, the residuals vs leverage plot does not reveal any outliers significant enough to heavily influence the model, as all points fall within the indicator lines (indicator lines are not able to be seen, suggesting that points fall within them). All of these factors indicate that a linear model is appropriate for this data.

d. Communicating results

To evaluate the relationship between ocean temperature and giant kelp frond elongation rate, a simple linear regression was used to test whether temperature could predict kelp growth.

The results suggest to reject the null hypothesis because a strong negative relationship was found between ocean temperature and giant kelp frond elongation rate, meaning kelp grew more slowly at higher temperatures (linear regression, $F(1, 30) = 26.93$, $R^2 = 0.473$, $p < 0.001$, $\alpha = 0.05$). On average, kelp elongation rate decreased by about 0.43 cm day^{-1} for every 1°C increase in ocean temperature.

e. Test implications

These results suggest that warmer ocean temperature ($^{\circ}\text{C}$) could reduce giant kelp frond elongation rate (cm day^{-1}). Because giant kelp forests provide crucial habitat and food resources for marine animals, a slower growth rate of these giant kelp fronds could result in resource availability for the dependent species. As the ongoing climate crisis worsens, ocean temperatures are projected to continue rising, which can further encourage the slowing of giant kelp frond elongation. Furthermore, it is crucial to continue investing time and funding into developing solutions to rising sea temperatures.

f. Double checking using a different test

```
# pearson correlation test to assess strength and direction of relationship
cor.test(
  # first variable: temperature from kelp data frame
  kelp$temp_c,
  # second variable: elongation rate from kelp data frame
  kelp$kelp_elong,
  # specification that it is a pearson correlation test
  method = "pearson"
)
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

The Pearson correlation test and the simple linear regression produced the same overall conclusion: to reject the null hypothesis in favor of accepting the alternative hypothesis, as both tests found a statistically significant negative relationship between ocean temperature and giant kelp frond elongation rate.

The Pearson correlation measured the strength and direction of the relationship between temperature and kelp elongation rate ($r = -0.688$), while the linear regression estimated the

amount of change in kelp growth associated with temperature (kelp elongation rate decreased by approximately 0.43 cm day^{-1} for every 1°C increase in ocean temperature).

Problem 2: Personal data

a. Updating my visualizations

Figure 1:

```
# creating a new object specifically for my cleaned data
my_data_clean <- personal_data |>
# cleaning up the names to include lowercase letters and underscores
janitor::clean_names() |>
filter(!is.na(temperature_h_w_c),
       temperature_h_w_c != "") |>
mutate(temperature_h_w_c = factor(temperature_h_w_c,
                                levels = c("H", "W", "C"), # order in which
                                ↪ levels will appear on plot
                                labels = c("Hot", "Warm", "Cold"))) # full
                                ↪ words for plot labels
```

```
# base layer ggplot with my_data_clean data frame
ggplot(data = my_data_clean,
       # setting x and y axes
       mapping = aes(x = temperature_h_w_c, # x axis temperature
                     y = water_intake_l, # y axis water intake in liters
                     color = temperature_h_w_c)) + # color by temperature
# first layer geom_boxplot to assess median amounts of water drank
geom_boxplot() +
# setting nice colors for my temperature values (intuitive ones)
scale_color_manual(values = c("Cold" = "skyblue", # blue for cold
                              "Warm" = "palevioletred", # this color is
                              ↪ just warm to me idk
                              "Hot" = "red3")) + # red for hot, duh
# renaming axes to be neater and more clear
labs(
  x = "Temperature: hot, warm, or cold", # x axis looking at temperature
  y = "Water intake (L)", # y axis looking at water intake
  # title that says something about the figure
  title = "On average, I drink more water when it is hot outside",
  # subtitle stating when the most recent data collection was
```



```

    subtitle = "Most recent data collected 5/25/2026"
  ) +
  # my favorite theme
  theme_bw() +
  # removing the legend, not super necessary here
  theme(legend.position = "none",
        # removing grid panels
        panel.grid = element_blank())

```

On average, I drink more water when it is hot outside

Most recent data collected 5/25/2026

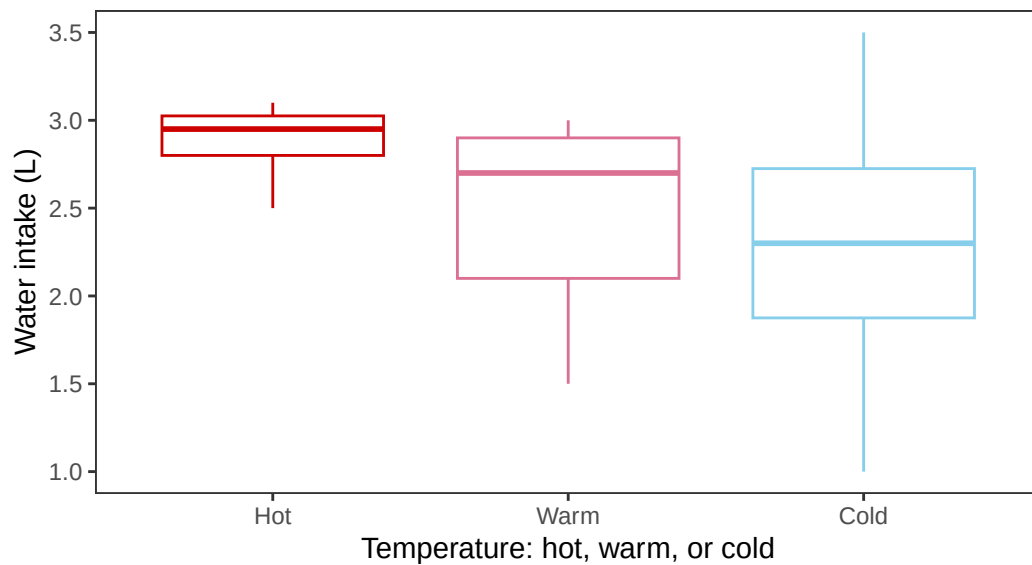


Figure 2:

```

# making new object specifically for this part, as i do not want to mess with
  ↳ my other cleaned data object in case i want to re-run
my_data_steps <- personal_data |>
  # cleaning names to be lowercase and include underscores
  janitor::clean_names() |>
  # dropping n/a values for duration of shift to filter for only days i
  ↳ worked
  drop_na(duration_of_shift_hrs)

```

```

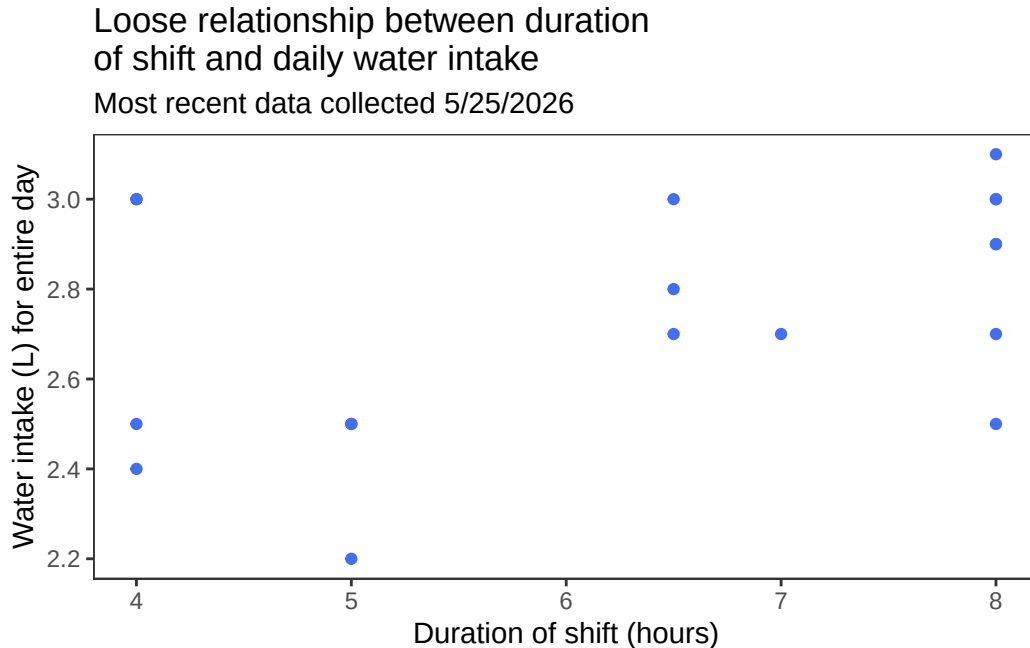
# base layer ggplot with my_data_steps data frame
ggplot(data = my_data_steps,

```

```

# setting x and y axes
mapping = aes(x = duration_of_shift_hrs, # x axis is duration of shift
              y = water_intake_l)) + # y axis is water intake
# first layer geom_point in pretty royal blue color
geom_point(color = "royalblue2") +
# setting axes, title, and subtitles
labs(
  # x axis title duration of shift
  x = "Duration of shift (hours)",
  # y axis title water intake
  y = "Water intake (L) for entire day",
  # bland statement for title because this relationship is kind of a
  ↪ stretch
  title = "Loose relationship between duration \nof shift and daily water
  ↪ intake",
  # subtitle most recent data collected on may 25
  subtitle = "Most recent data collected 5/25/2026"
) +
# voted best theme on a reddit post i read
theme_bw() +
# getting rid of panel grid
theme(panel.grid = element_blank())

```



b. Creating captions for figures

Figure 1: Relationship between daily water intake (L) and daily temperature (hot, warm, cold)

On average, my median daily water intake (measured in liters) is higher when I experience warmer temperatures that day, as shown by the median bar for the data in red (hot) being higher than the median bar for the data in pink (warm) and the median bar for the data in blue (cold).

Figure 2: Relationship between duration of shift (hrs) and daily water intake (L)

There is not a strong correlation between the duration of the shift worked that day (measured in hours) and daily water intake (measured in liters). Based on the orientation of the blue points, which represent individual, daily observations, there may be a positive relationship between these two variables, but a large amount of outliers make it difficult to discern a concrete relationship.

Problem 3: Affective visualization

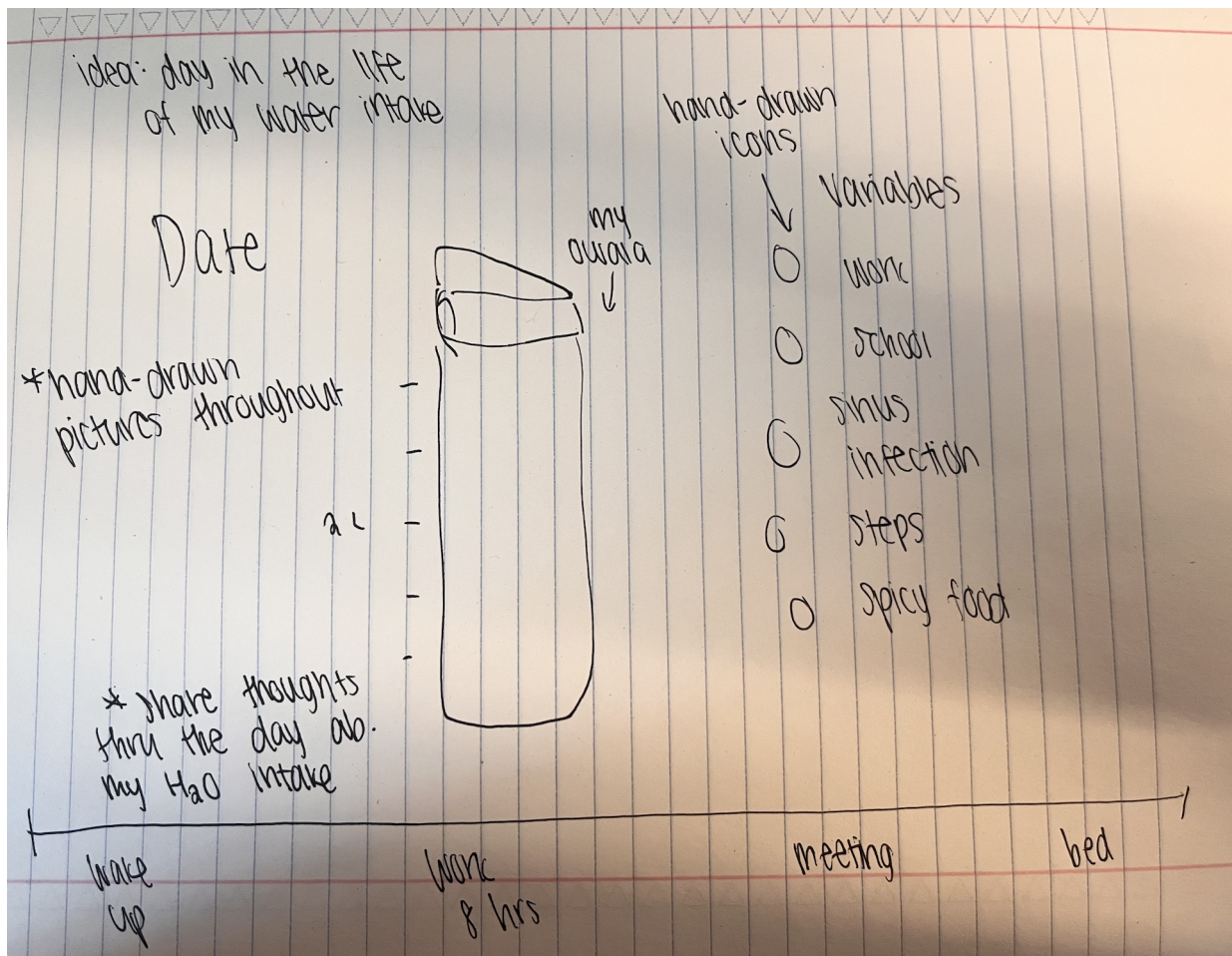
a. Describing my affective visualization

For my affective visualization, I want to utilize the fact that I am recording a ton of variables every day, including whether or not I went to school, work, if I had a sinus infection, and much more. Consequently, for my affective visualization, I would like to use my water consumption for a day to tell a story about my entire day, like a “day-in-my-life” premise. I want it to have character and reflect my personal artistic style because it is *about me*. I am thinking of drawing my Owala water bottle as the main component and visually show my water consumption, then surround this with different variable explanations about what I did that day that could have contributed to my water consumption.

b. Creating a sketch on paper of my idea

Paper sketch:

```
# code to include screenshot of affective visualization paper sketch
knitr::include_graphics(
  ↪ "drawings_and_rough_visualizations/stats_paper_drawing.png")
```



Note: Sorry, my lighting is not the best!

c. Making a draft of my visualization

Rough draft:

```
# code to include screenshot of affective visualization rough draft
knitr::include_graphics(
  ↪ "drawings_and_rough_visualizations/stats_visualization.png")
```

Ella's

daily
H₂O
intake

May 15

Today's temp:

COLD + GLOOMY

Today's variables:



work: yes



school: yes



sinus infection:
yes (always)



steps:
12,071



spicy food:
2x



my throat
hurts!

rushing, but
staying hydrated!

my throat
hurts...
worse?

CHUG!

REFILL!

REFILL!

CHUG!

Woke up
sick

8 hr
housekeeping
shift



research
meeting



Go to
bed...
sick

d. Beginning to form an artist statement

- For this piece, I am showing my daily water intake (L), and what conditions I experienced that day that may influence it.
- After looking over the resources provided to us, I really enjoyed the character in Stefanie Posavec and Giorgia Lupi's Dear Data project. I feel as though, in this project, they showed a lot of character, which inspired me to include many personal aspects in my work, such as my thoughts throughout the day, funny little remarks, and other small components.
- I decided to do a digital drawing because I felt as though hand-drawing my icons the way that I experience and "feel" my variables would be the most authentic. For example, my hand-drawn sad "sinus infection" face is exactly how I feel when I have a sinus infection.
- I created my work by first thinking about what I wanted to accomplish, which was telling a story about my day and mood through my work. So, I started with a time-line to outline my day and what my thoughts were about water. I added my variables as well, and then included lots of little icons that I associate with those variables.

e. Hyperlink to slideshow for 5/29 workshop

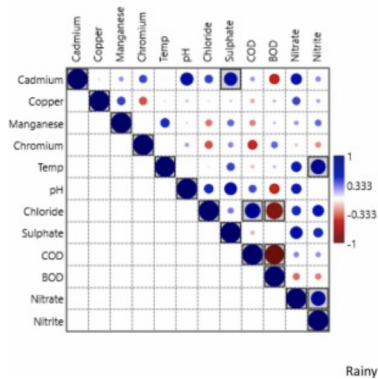
[Click here](#) to view slideshow presentation

Problem 4: Statistical critique

a. Revisiting and summarizing

For my statistical critique, I am examining a paper entitled, "Behavioural dynamics of water quality parameters and human exposure assessment of trace metals in Agulu Lake Water, South-East Nigeria" from the ScienceDirect journal. In this paper, they include a Spearman correlation, as they are focusing on examining relationships between pairs of variables. In this scenario, the response variable could be one water quality parameter (like concentration of a specific trace metal) and the predictor variable could be another water quality parameter (like another trace metal).

```
# code to include png of the spearman correlation figure from paper
knitr::include_graphics("data/spearman_screenshot copy.png")
```



b. Assessing visual clarity

I do think that the authors clearly represented their statistics in figures. Because they had so many trace metals (and relationships between these trace metals) to analyze, I believe this figure is an efficient way to visually demonstrate the correlation between two trace metals. This figure does not show explicit values per circle (Spearman correlation values), but the relative size of the circle compared to other sizes makes it easy to tell whether or not the presence of two trace metals are heavily related. The addition of the scale on the side also makes it very clear.

c. Assessing aesthetic clarity

This figure is mostly ink, as they data is represented by purposefully sized dots. Given how many relationships between trace metals this figure is exploring, I feel as though they have excellent aesthetic clarity with minimal visual clutter. This figure shows 72 (half the figure is filled because the other half would be repetitive) relationships, so I commend them on their creative figure creation to convey so much data in one figure.

d. Recommendations

The only recommendation I would have for the authors is to add a title disclosing that the figure shows the correlation between the two types of trace metals found in the water. Although it is relatively clear that is the intended purpose of the figure in the caption, I think a title would serve this figure well because it is often the first component of a figure that a reader looks at is the title, and it lets them know what to expect while analyzing the figure. It is my understanding that a great figure is at least relatively clear to readers with just the title and figure itself.